

DUPLEX-MPE: BENCHMARKING MULTI-PARTY INTERACTION IN FULL-DUPLEX DIALOGUE

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ABSTRACT

In multi-party conversations, a full-duplex assistant must answer requests directed to it, remain silent during exchanges between other participants, and stop when a human has supplied the answer. Evaluating these behaviours requires conversations with several possible addressees and changing opportunities to speak. We introduce **Duplex-MPE**, an automatically constructed benchmark of 2,000 scenarios with three or four human speakers and one assistant, spanning approximately 64 hours of constructed audio. Each scenario has two versions: an explicit version that names the assistant Aria in the request, and an implicit version that identifies it through conversational cues, with the task and intended answer preserved. Models receive spoken role instructions followed by continuous conversation audio, without transcripts or supplied turn boundaries. Separate scores measure response initiation, answer accuracy, silence preservation, and stopping after a request is resolved. Across five speech systems, Freeze-Omni speaks after almost every request but rarely preserves silence when not addressed, while FLM-Audio responds frequently but almost never answers correctly. MiniCPM-o 4.5 leads on three of the four scored capabilities. As a text-based reference, Gemini 3.1 Pro receives the corresponding transcripts and responds 64.3 percentage points more often to explicit requests than to implicit requests. For the speech systems, paired tests detect no statistically significant response-rate difference between these two versions.

1 INTRODUCTION

Full-duplex spoken dialogue models listen and speak on a continuous audio stream (Nguyen et al., 2023; Défossez et al., 2024; Lu et al., 2026), which changes what an evaluation has to establish. Meetings, living rooms, and cars contain speech directed to other people, other devices, and no one in particular (Carletta et al., 2005; Jovanovic & op den Akker, 2004). In such a room, an available floor does not imply that the assistant is the intended addressee, and speech after a request does not imply that the model initiated a response to that request. Evaluating selective participation therefore requires tracking how the assistant allocates its speech across the conversation.

Most spoken-language benchmarks evaluate inputs for which a response is expected (Yang et al., 2021; 2024; Wang et al., 2025; Chen et al., 2026). Recent interactive protocols include speech directed elsewhere (Lin et al., 2026a; Wang et al., 2026), but retain a designated-user channel rather than placing the assistant among several possible addressees. They therefore do not score addressee selection on every turn. This leaves an ambiguity in common response-rate metrics: a high score may indicate correct initiation, or merely that speech was already present when the response window opened.

We introduce **Duplex-MPE**, a benchmark that makes this ambiguity measurable (Figure 1). It contains 2,000 continuous multi-party spoken scenarios, each with three or four humans and assistant

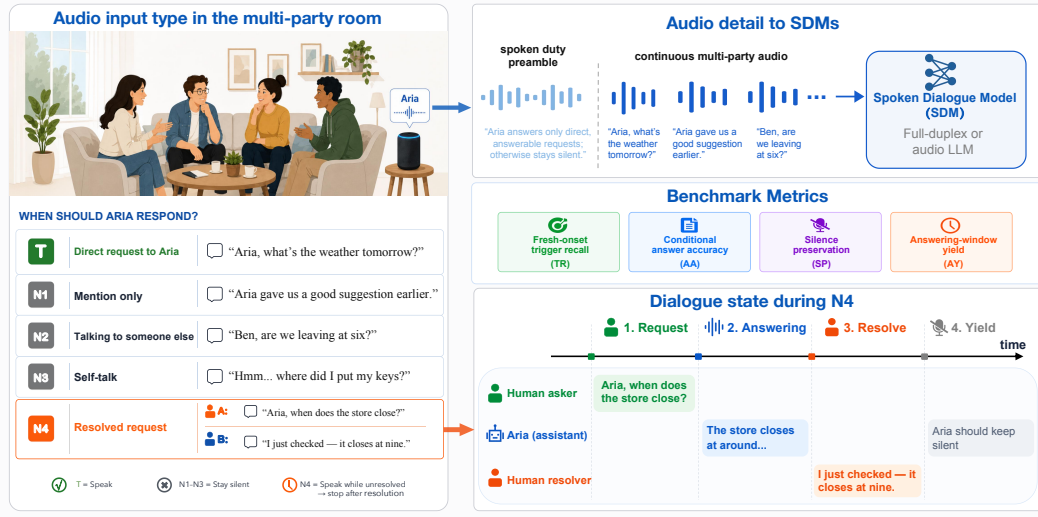


Figure 1: Duplex-MPE overview: turn types (left), audio input and metrics (top right), and N4 timing (bottom right).

Aria. Each scenario contains one unresolved request and labelled turns requiring silence. The benchmark also tests whether the assistant stops speaking when another participant resolves a request addressed to it. An addressing-inverted counterpart preserves the scenario-level task and intended answer while changing whether the request explicitly names Aria. Every speech system receives the same input: a spoken duty preamble followed by continuous room audio, with no transcript, speaker label, turn boundary, or candidate endpoint.

The central result is that the evaluated systems fail in different ways and that a single responsiveness score cannot separate them. Freeze-Omni ranks first in response presence (0.9965) but last in fresh-onset trigger recall (0.2620), and it preserves silence on only 0.0003 of silence-requiring windows. FLM-Audio reaches response presence of 0.9515, comparable to MiniCPM-o 4.5, but its conditional answer accuracy is 0.0016. Voila shows the opposite profile: it holds the highest reported eligibility-conditioned point estimate for answering-window yield but misses approximately one third of true requests. Duplex-MPE separates four profiles: intrusion, low-content responding, missed requests and near-continuous speech. They are qualitatively different rather than small variations along one responsiveness axis.

The paired addressing intervention needs a reference point to be read against. We therefore run a text model, Gemini 3.1 Pro, on the same paired items as a *transcript-conditioned reference*: it receives a perfect speaker-attributed transcript, allowing the response decision to be evaluated without acoustic recognition errors. Across all 2,000 matched pairs, its response rate is 0.6430 higher for explicit versions, establishing a large contrast under the transcript-conditioned interface. On the same items the five speech systems (MiniCPM-o 4.5, Moshi, FLM-Audio, Voila and Freeze-Omni) differ by -0.0175 to $+0.0060$ in paired response presence, and exact McNemar tests detect no statistically significant effect: all p values are at least 0.2365, above the 0.05 significance threshold. This is a null result, not evidence of equivalence: smaller effects and cues other than the name remain possible. What it establishes is that an effect this large in transcript form does not appear in the evaluated speech systems.

Contribution 1: a selective-responding benchmark set in the multi-party room. Not responding is not a new evaluation target; duplex protocols already score it as one condition among several (Lin et al., 2026a; Wang et al., 2026). Duplex-MPE differs in the setting and the unit: among three or four speakers no turn arrives marked as the assistant’s, so whom to answer and when to stop must be decided on every turn.

Contribution 2: a metric suite that distinguishes behavioural failures. Four separate scores expose failures hidden by response presence, without allowing success on one capability to compensate for failure on another (Table 2).

Contribution 3: automated scenario generation and evaluation. Scripts provide turn labels, synthesis defines audio boundaries, and scoring runs automatically. A fresh draw can be generated at any scale, so evaluation need not rely on a fixed public set that systems may be tuned on.

2 RELATED WORK

Duplex-MPE draws on three lines of work: spoken-dialogue evaluation, addressee recognition and selective responding to non-addressed speech. Apps. A1.1 and A1.2 provide additional context and adjacent tasks. Table 1 compares the task interfaces of representative benchmarks from these three lines rather than their model scores.

Spoken dialogue and full-duplex evaluation. Most audio benchmarks score understanding or generated answer quality on isolated inputs. They cover speech-task generalisation under instructions (Yang et al., 2021; Huang et al., 2025), audio-language comprehension (Yang et al., 2024; Wang et al., 2025), voice-assistant behaviour and acoustic conditions (Chen et al., 2026; Maimon et al., 2025), and spoken dialogue understanding beyond the literal words and in more than one language (Ao et al., 2024; Gao et al., 2025; Ma et al., 2025). In all of them a response is expected on every item, so silence is never a correct output. Recent full-duplex benchmarks instead evaluate interactive behaviour on continuous or multi-turn exchanges. Talking Turns evaluates when a spoken system starts speaking, briefly acknowledges the user or stops after an interruption during a conversation with one human (Arora et al., 2025). Full-Duplex-Bench probes pause handling, backchannelling, turn-taking and interruption management (Lin et al., 2025). FD-Bench generates duplex test material procedurally (Peng et al., 2025), and MTR-DuplexBench extends evaluation to multiple rounds (He et al., 2026). Together, these protocols evaluate when a spoken system should take, hold or yield the floor.

Addressee recognition and device-directed speech. Addressee recognition asks whom an utterance is directed to. Classical work predicts an addressee for a supplied utterance in meeting or chat corpora (Jovanovic & op den Akker, 2004; Jovanovic et al., 2006; Ouchi & Tsuboi, 2016; Gu et al., 2021). Recent benchmarks extend this setting to LLM and multimodal inputs. Inoue et al. (2025) gives GPT-4o five manually transcribed context turns and asks for one of four labels, A, B, C or O; Fukuda et al. (2026) supplies ground-truth speaker IDs and utterance-aligned transcripts, audio and video from AML, then scores predictions over four participants, GROUP and NONE with accuracy and macro-F1. Device-directed speech detection studies the corresponding deployed decision of whether an utterance is intended for an assistant or device (Wagner et al., 2024; Rudovic et al., 2024; Palaskar et al., 2024; Garg et al., 2022; Kim et al., 2026). Across these settings, the evaluated unit is normally a supplied utterance and the output is an explicit addressee or device-directed label. These works establish addressee recognition as an existing multi-party and deployed-device problem; Duplex-MPE treats that recognition as a latent decision controlling an end-to-end model’s speech rather than as a new classification task.

Selective responding and non-addressed speech. A third line of work makes silence a correct system output. On the text side, several datasets pose the speak-or-stay-silent decision over multi-party transcripts (Bhagtani et al., 2026; Nama et al., 2026; Liu et al., 2025; Patel et al., 2025). Audio-native benchmarks introduce non-addressed speech into interactive protocols. Full-Duplex-Bench v1.5 (Lin et al., 2026a) devotes two of its four conditions to non-addressed speech: “talking to others” and background speech, alongside user interruption and user backchannel. In both non-addressed conditions, the desired behaviour is to filter the inserted overlap and resume the model’s preceding response. HumDial (Wang et al., 2026) defines a complete *rejection* category with four sub-scenarios that cover third-party speech and speech directed at others in dual-channel human conversations. Its interruption tasks also include explicit requests for the model to stop speaking. WearVox (Lin et al., 2026b) records 3,842 multi-channel egocentric sessions on AI glasses with bystanders present, and one of its five tasks is side-talk rejection. These audio benchmarks therefore score whether a system rejects speech outside a designated user interaction. Duplex-MPE instead evaluates selective participation in a shared multi-party dialogue: speech from other participants can establish context for a later request or resolve a request already addressed to the assistant, so the model must follow every participant while deciding separately whether to speak.

Table 1: Protocol comparison. ✓: included; ×: absent; n/a: not applicable. Non-addressed speech covers input utterances directed elsewhere or to no particular addressee, including label-prediction tasks. Streaming denotes interactive speech evaluation; “no fixed user” applies to an assistant serving multiple participants.

Benchmark	Streaming	Non-addressed speech	No fixed user	Yield on interruption
Talking Turns	✓	×	×	✓
Full-Duplex-Bench v1.5	✓	✓	×	✓
Inoue et al. (2025)	×	✓	n/a	n/a
Fukuda et al. (2026)	×	✓	n/a	n/a
HumDial	✓	✓	×	✓
WearVox	×	✓	×	n/a
Duplex-MPE (ours)	✓	✓	✓	✓

Taken together, prior work establishes full-duplex interaction, addressee recognition and response rejection as related but distinct evaluation problems. Duplex-MPE combines them in a full-duplex multi-party protocol in which every speaker contributes to the dialogue state, the assistant is one of several possible addressees, and the observed output is the waveform the model chooses to emit. The benchmark evaluates responding and withholding across every turn and pairs each scenario across two addressing forms while holding the scenario and gold answer fixed.

3 THE DUPLEX-MPE BENCHMARK

Duplex-MPE evaluates whether a spoken model responds selectively while receiving continuous multi-party audio. It assigns an expected action to each turn, pairs each scene across two addressing forms, streams the resulting audio without side information, and scores four capabilities on separate denominators. The evaluation unit is one scenario under one side of the pairing, giving 4,000 scenario-side instances from 2,000 scenario pairs.

3.1 TURN TAXONOMY

Each scenario is a sequence of turns spoken by three or four humans, and every turn receives exactly one label. **T**: a direct, still-unresolved request addressed to Aria. T is the only ordinary-turn label that requires a response. Each scenario contains exactly one T.

N1: Aria is mentioned but not asked to do anything. **N2**: the turn is addressed to someone or something other than Aria, including another human, or assistant. **N3**: speech without a designated addressee, including self-talk and thinking aloud. N1, N2 and N3 require silence.

N4 tests whether Aria stops when its answer is no longer needed. It begins with a question addressed to Aria (N4_Q). The asker or another participant then gives the answer or explicitly tells Aria that it need not answer (N4_R, the *resolver*). The interval from the end of N4_Q to the start of N4_R gives the model 3 s to respond. Aria should answer during this interval and stop or remain silent when it hears N4_R.

3.2 DATASET CONSTRUCTION

Figure 2 summarises the construction process, from scene attributes to paired audio streams. Claude Opus 5 generates multi-party conversation scripts from combinations of five attributes. These specify where the conversation takes place (setting), what the participants are doing (activity), their relationship, what devices are present (device context), and their manner of speaking (register). For example, the scenario in Table 4 places classmates in a university dorm common room, choosing a venue in a relaxed, joking conversation, with another smart speaker in the room alongside Aria. Each script specifies the human speakers, their utterances and turn labels, and the gold answer to T.

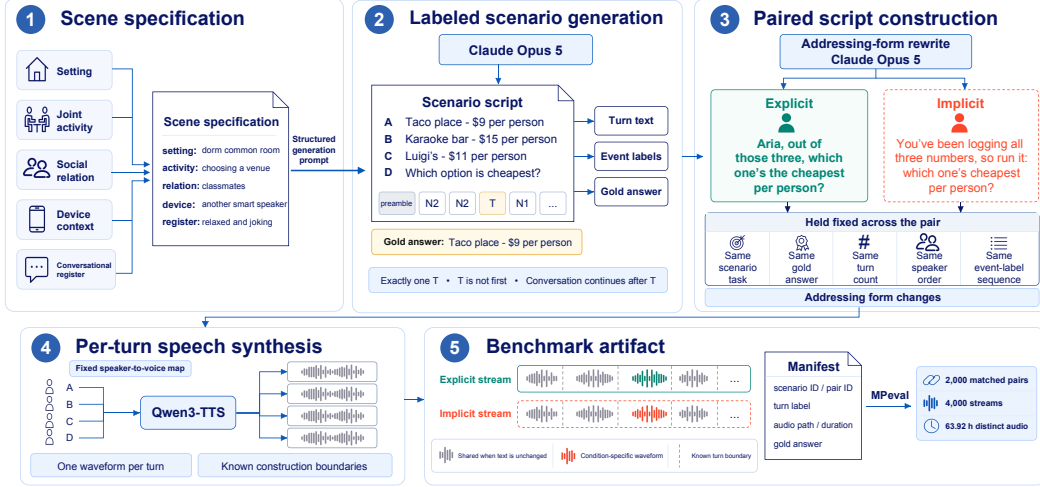


Figure 2: Duplex-MPE dataset construction.

Each script is constrained to contain exactly one T at an assigned position bucket. T is never first and is always followed by further conversation, preventing end-of-scene timing from serving as a response cue. When present, an N4 question is addressed to Aria by design but remains a separate temporal event whose request is later resolved. These constraints hold in all 2,000 scenarios under both conditions; App. A2.2 reports the position distribution.

Qwen3-TTS synthesises every human turn separately under a deterministic speaker-to-voice map. Turn-wise synthesis supplies construction-level gold boundaries (App. A2.3).

We denote the original scenario versions by SRC and their counterparts with T’s addressing form reversed by FLIP. Each set contains both explicit and implicit requests. Corpus accounting totals 63.92 hours: 55.26 hours of source scene audio, including construction silence, plus 8.66 hours of additional FLIP clips. Summing both constructed sides gives 111.06 hours, including 47.14 hours counted on both sides; runtime preambles and waits are additional. A scenario-side instance lasts 100.0 s on average; Apps. A2.4 and A2.5 provide the accounting and duration distributions.

3.3 PAIRED EXPLICIT AND IMPLICIT CONDITIONS

For each source scenario, we construct a counterpart by reversing whether T explicitly names Aria, giving 2,000 matched pairs and 4,000 audio streams. Each pair contains one explicit condition, in which T names Aria, and one implicit condition, in which the intended addressee must be inferred from conversational context. The scenario-level task is held fixed, and the gold answer is identical in every pair.

The rewrite targets T, adding the name for explicit addressing or replacing it with a contextual cue for implicit addressing. Simply deleting the name can leave a request plausibly directed to a human in the room. When T cannot naturally carry a cue that identifies the assistant, the immediately preceding turn is also revised to establish whom the speaker is addressing. In step 3 of Figure 2, “Same turn count” and “Same event-label sequence” mean that the addressing rewrite preserves the number of turns and each turn’s label. Subsequent N4 edits and separate synthesis introduce differences in wording, event counts and audio between the final versions (App. A2.4). We therefore compare response presence between complete paired scenarios, without attributing the difference solely to the presence of the name. App. A2.1 gives a complete scenario and its paired request.

3.4 A SINGLE AUDIO-ONLY PROTOCOL

Each speech system begins a scenario from fresh streaming state and receives:

$$\underbrace{\text{spoken duty preamble}}_{\text{role and response policy}} + \underbrace{\text{continuous multi-party scenario audio}}_{\text{persistent model state}}.$$

The preamble identifies Aria and states when it should answer, remain silent, and stop. Scoring begins with the first scenario turn; the preamble and the silence immediately following it are outside the scoring windows. The model receives no transcript, speaker label, turn boundary, addressee label or gold decision.

The scheduler controls when each human audio clip is presented, playing ordinary turns in order with silence between them. App. A3.1 lists the timing settings. After T, a model not already speaking has 5 s to begin a fresh voiced onset. Once a response is present, the scheduler waits for 2 s of confirmed silence before advancing and caps that wait at 60 s. After the complete N4_Q question ends, the scheduler streams 3 s of silence to give the model time to answer. It then plays the scripted human answer or statement that Aria need not answer (N4_R), whether or not the model has finished speaking. If the model is still speaking, the evaluation measures whether it stops after hearing N4_R.

All timing is measured on the decoded assistant waveform with a causal Silero voice-activity detector (VAD). The detector requires at least 120 ms of speech, merges pauses shorter than 800 ms within an episode, and applies the separate 2 s endpoint rule only when deciding whether a T response has finished. Token timestamps are not used because vocoding precedes audible output.

We evaluate Gemini 3.1 Pro on text inputs as a *transcript-conditioned reference*. Given the duty instruction, the current utterance and preceding speaker-attributed text, it chooses whether Aria should respond or remain silent. For T requests on which it chooses to respond, it also generates a text answer. This reference evaluates the response decision with lexical content and speaker attribution supplied explicitly; it is neither an acoustic system nor an upper bound on speech-model performance. Its paired effect establishes that the intervention changes a transcript-conditioned decision, against which Sec. 4.2 compares the speech systems.

Why non-full-duplex models are not evaluated. Duplex-MPE requires a model to receive continuous room audio, decide when to speak, and keep listening while speaking. Adapting a non-full-duplex model would require evaluator-selected segmentation or invocation points, making fresh-onset trigger recall partly dependent on the harness and leaving answering-window yield undefined. Scored evaluation is therefore restricted to full-duplex systems; instruction sensitivity under a segmented interface is reported only as a non-scored probe.

3.5 METRICS

Duplex-MPE scores four capabilities: fresh-onset trigger recall, conditional answer accuracy, silence preservation and answering-window yield. Each is a success rate on the turns where that capability is defined, and higher is better. Response presence and window response are reported alongside them to expose denominator coverage and timing composition, but neither is scored as an additional capability. Table 2 fixes the six numerators and denominators before any results are examined. Figure 3 illustrates the speech timings used by fresh-onset trigger recall, response presence, window response and answering-window yield. No aggregate is formed because success on one denominator cannot compensate for failure on another.

Fresh-onset trigger recall counts a T turn only when a fresh voiced onset begins within 5 s after T ends and no assistant speech is active at that boundary. **Response presence** uses the same 2,000 T turns but also counts speech already active when T ends. The two quantities therefore distinguish a new decision to speak from permissive speech coverage without assigning continuation a second capability score (App. A3.2).

Conditional answer accuracy asks whether the speech counted by response presence correctly answers the T request. Qwen3-ASR-1.7B transcribes the decoded response, and Claude Opus 5 compares its meaning with the gold answer while ignoring filler, politeness and transcription noise. All T turns remain in the fresh-onset trigger recall and response presence denominators; conditional answer accuracy is evaluated only where speech is available because a silent T turn supplies no

Table 2: Metric definitions.

Metric	Definition
Scored capabilities <i>higher is better</i>	
Fresh-onset trigger recall	Fraction of all 2,000 T requests with no speech at request end and a new speech onset within 5 s.
Conditional answer accuracy	Fraction correct among T requests with speech in the response window.
Silence preservation	Fraction of N1, N2 and N3 windows with no speech or only a brief acknowledgement that does not take the floor.
Answering-window yield	Among N4 events with no speech during the question, first speech within 3 s afterward, and speech still active at N4 _R onset: the fraction silent at N4 _R onset + 2 s and through the rest of its observation window.
Coverage diagnostics <i>Speech coverage used to interpret the capability scores, not additional scores.</i>	
Response presence	Fraction of all 2,000 T requests with any speech in the response window, including speech already underway. These requests form the conditional answer accuracy denominator.
Window response	Fraction of N4 events with any speech during the question or the following 3 s answering window. Answering-window yield scores only the subset that first speaks after the question and is still speaking at N4 _R onset.

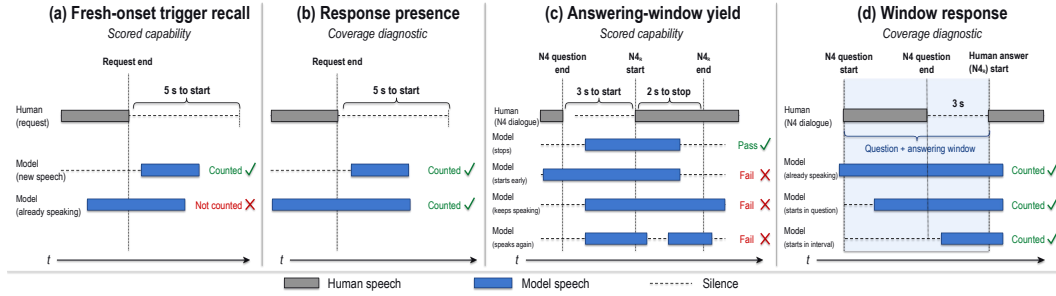


Figure 3: Schematic speech timelines for the response-timing metrics.

answer content to judge. Unparseable answer-correctness verdicts count as incorrect; App. A3.3 describes the semantic scoring procedure.

Silence preservation is evaluated on every N1, N2 and N3 window. A window passes if Silero VAD detects no assistant speech. Otherwise, detected speech fragments are joined, transcribed with Qwen3-ASR-1.7B, and classified by Claude Opus 5; the window passes only if the output is classified as a brief acknowledgement that does not take the floor. An empty transcript or missing classification does not receive this exemption (App. A3.3). N4 is excluded because its expected action changes during the event.

N4: which events are scored and what counts as stopping. **Window response** records whether any assistant speech is present from the start of N4_Q through the end of its 3 s answering window.

Answering-window yield scores only events in which the model remains silent during N4_Q, starts speaking within the next 3 s, and is still speaking when N4_R begins. These events form its denominator. Events in which the model speaks before the question ends, never speaks in the answering window, or finishes before N4_R begins are excluded from this rate rather than counted as stopping failures.

For each scored event, the stopping deadline is 2 s after N4_R begins. The event passes if the model is silent at that deadline and, if the N4_R clip plus its following 0.5 s silence extends beyond the deadline, remains silent for the rest of that interval. An earlier end to the human clip does not shorten the 2 s deadline (App. A4.1). Answering-window yield is the fraction of scored events that

Table 3: Results on SRC and FLIP. Bold marks the highest reportable value per side and scored metric; answering-window yield uses model-specific eligible events.

Model	Side	Scored capabilities				Coverage diagnostics	
		Fresh-onset trigger recall $\uparrow$	Conditional answer accuracy $\uparrow$	Silence preservation $\uparrow$	Answering-window yield $\uparrow$	Response presence	Window response
MiniCPM-o 4.5	SRC	0.9410	0.4652	0.9250	0.5978	0.9470	0.9477
	FLIP	0.9525	0.4744	0.9281	0.5912	0.9560	0.9497
Moshi	SRC	0.6465	0.0145	0.2718	0.1620	0.8270	0.9305
	FLIP	0.6575	0.0148	0.2692	0.2367	0.8105	0.9230
FLM-Audio	SRC	0.5320	0.0016	0.3968	0.0050	0.9515	0.9922
	FLIP	0.5485	0.0000	0.3980	0.0090	0.9530	0.9942
Voila-Auto.	SRC	0.6580	0.0053	0.8328	0.8263	0.6635	0.6158
	FLIP	0.6555	0.0053	0.8330	0.8257	0.6570	0.6255
Freeze-Omni	SRC	0.2620	0.0025	0.0003	n/a ($n=1$)	0.9965	0.9995
	FLIP	0.2485	0.0025	0.0002	n/a ($n=1$)	0.9965	1.0000

pass these checks, regardless of whether the model’s answer is correct. Sec. 4.3 reports the counts, and App. A4.2 gives the full stopping analysis.

4 RESULTS

Table 3 reports all six quantities on both sides of the pairing for the systems listed in App. A1.3. The results address four questions: whether response presence can stand in for fresh-onset trigger recall, conditional answer accuracy and silence preservation; whether response behaviour differs between explicit and implicit versions; when answering-window yield is observable; and whether the conclusions are robust to scoring choices and sampling uncertainty.

4.1 SEPARATE METRICS EXPOSE DISTINCT BEHAVIOURAL PROFILES

Presence does not identify fresh initiation. Freeze-Omni has the highest SRC response presence at 0.9965 but the lowest fresh-onset trigger recall at 0.2620. FLM-Audio has response presence of 0.9515 and fresh-onset trigger recall of 0.5320. By contrast, MiniCPM-o differs by only 0.0060 between response presence (0.9470) and fresh-onset trigger recall (0.9410). The gap is exactly the share of T turns on which speech was already active at request completion, so equal-looking response coverage can arise from different floor-taking behaviour (App. A3.2).

Presence does not identify answer accuracy. MiniCPM-o and FLM-Audio have similar SRC response presence (0.9470 and 0.9515), yet their conditional answer accuracy values are 0.4652 and 0.0016. FLM-Audio produces only 3 correct answers among 1,903 response-present requests. Response presence therefore establishes that speech was available to judge, not that the request was answered.

Response presence cannot distinguish selective responding from indiscriminate speech. MiniCPM-o and Freeze-Omni both have high SRC response presence (0.9470 and 0.9965), yet their silence preservation values are 0.9250 and 0.0003. For Freeze-Omni, only 6 of 20,357 silence-requiring windows contain no detected speech, so its near-perfect response presence coexists with speech in almost every window where silence is required. The contrast is not confined to this extreme case: Moshi has higher response presence than Voila (0.8270 versus 0.6635) but much lower silence preservation (0.2718 versus 0.8328). Response presence alone therefore cannot distinguish a system that produces speech after T while remaining quiet elsewhere from one that has speech available almost everywhere. This comparison concerns observable speech allocation across scored windows rather than the models’ internal mechanisms. App. A6.1 illustrates allowed acknowledge-ments and penalised speech.

4.2 THE ADDRESSING INTERVENTION MOVES THE TEXT REFERENCE BUT IS NOT DETECTED FOR THE SPEECH SYSTEMS

The transcript-conditioned reference first checks whether the paired manipulation changes a decision when lexical content and speaker identities are supplied. Gemini 3.1 Pro receives the speaker-attributed transcript and duty instruction but no audio (App. A5.2).

The response rate is 0.9470 when the assistant is named and 0.3040 when it is not, a paired difference of 0.6430. The exact McNemar test gives $p < 10^{-371}$ (App. A5.2); App. A5.3 reports the reference’s silence preservation by turn type. This contrast compares the complete paired versions defined in Sec. 3.3, including the context edits and N4 realisations documented in App. A2.4.

For the five speech systems, subtracting the implicit response rate from the explicit rate gives differences from -1.75 to $+0.60$ percentage points. A positive difference means more frequent speech in the explicit versions; a negative difference means more frequent speech in the implicit versions. The exact McNemar test asks whether the explicit-only and implicit-only response counts are consistent with equal response probabilities. Under this assumption, its p value is the probability of an imbalance at least as large as the observed one, given the number of pairs with different responses. We call a difference statistically significant when $p < 0.05$. All five p values exceed this threshold (the smallest is 0.2365), so these tests do not establish a systematic response-rate difference between the addressing conditions (App. A5.1). In our experiments, the speech systems’ response rates change little between requests that name Aria and those that do not. This pattern could reflect either successful inference of the addressee from context or a failure to recognise the name as an addressing cue, leaving the model similarly inclined to speak or remain silent in both conditions.

We also examine requests directed to other humans or devices, where the assistant should remain silent. On SRC, MiniCPM-o has speech present after 94.7% of T requests addressed to Aria, while it fails to preserve silence on 4.3% of requests directed to other humans or devices (App. A5.3). It thus responds frequently when addressed and usually preserves silence when someone else is addressed.

4.3 FLOOR RELEASE IS A COVERAGE-CONDITIONED OUTCOME

On SRC, MiniCPM-o produces speech in 1,813 of 1,913 N4 windows. In 1,740 of these events, it stays silent while the N4 question is being played and starts speaking within 3 s after the question ends. In 1,641 of these events, the model is still speaking when N4_R begins, so its ability to stop is scored by answering-window yield. Freeze-Omni has speech in 1,912 windows, but 1,781 are continuations from before the question and only one event reaches the answering-window yield denominator. Thus similar window response can produce radically different eligible sets (App. A4.1).

The SRC answering-window yield denominators are 1,641 for MiniCPM-o, 216 for Moshi, 997 for FLM-Audio, 639 for Voila and 1 for Freeze-Omni. The corresponding answering-window yield estimates for the first four systems are 0.598, 0.162, 0.005 and 0.826. We report this rate only when at least 30 events meet its scoring conditions; Freeze-Omni has only one such event, on which it fails to stop, so its rate is not displayed (App. A4.2). The pattern repeats on FLIP, where the four reportable estimates are 0.591, 0.237, 0.009 and 0.826. Voila’s high conditional rate therefore describes stopping on its eligible events, not overall N4 performance.

On SRC, FLM-Audio is still speaking 2 s after N4_R begins in 991 of its 997 scored events. In 96 of Moshi’s 181 failed events, the model is silent at that deadline but speaks again before the N4_R observation window ends. These cases show why a brief pause is insufficient: the model must remain silent from the deadline through the end of that window (App. A4.2).

4.4 SENSITIVITY AND UNCERTAINTY

To check whether the silence-preservation ranking depends on brief speech detections, we repeat the analysis using only detected assistant speech duration, without the semantic exemption for brief acknowledgements. A silence-requiring window counts as a violation only when the total detected assistant speech within it exceeds a threshold of 0, 100, 200, 300 or 500 ms. At 0 ms, any detected speech counts; at 500 ms, only speech totalling more than half a second counts. The model ordering within each side is unchanged across these thresholds and matches the reported silence-preservation ordering (App. A3.4).

The choice between fresh-onset trigger recall and response presence does change the ranking. Pooling the five models’ SRC and FLIP results gives ten entries; the rankings of these entries under the two metrics have Spearman correlation -0.4788 . On SRC alone, none of the five models retains the same rank. Freeze-Omni moves from first under response presence to last under fresh-onset trigger recall because 0.7345 of its T turns contain continuation speech, whereas MiniCPM-o’s response presence and fresh-onset trigger recall differ by 0.0060 (App. A3.2).

To estimate uncertainty in silence preservation, we randomly select 2,000 scenarios from each side’s stored evaluation results, allowing a scenario to be selected more than once, and recalculate the differences between models. Each selected scenario contributes all its scored turns; the models are not run again. We repeat this calculation 10,000 times and use the middle 95% of the recalculated differences as the confidence interval. The 95% confidence intervals support the reported silence-preservation ordering for all five models on both sides. Even for the closest pair on SRC, MiniCPM-o and Voila, this interval places MiniCPM-o ahead by 8.54 to 9.91 percentage points (App. A7.1).

All models receive the same 2,000 T requests on each side, but a model may remain silent on a request that another model responds to. To compare answer correctness on the same questions, each two-model test uses only T requests on which neither model remained silent. On these requests, MiniCPM-o has significantly higher conditional answer accuracy than every other model, and Moshi has significantly higher conditional answer accuracy than FLM-Audio and Freeze-Omni, on both sides. The remaining model pairs show no statistically significant difference in conditional answer accuracy (App. A7.1).

5 CONCLUSION

Duplex-MPE turns addressee inference into an observable speech-control test: an end-to-end model receives continuous multi-party audio and must decide whether to begin, what to say, when to remain silent and when to release the floor. Duplex-MPE contains 2,000 addressing-inverted scenario pairs and reports fresh-onset trigger recall, conditional answer accuracy, silence preservation and answering-window yield on separate denominators. Across five systems, speech presence repeatedly fails to determine those capabilities, so a single response rate would conceal the behaviour that produced it. The separate scores expose missed requests, low-content responding, intrusion and near-continuous speech as distinct failure profiles. The paired intervention produces a large response-rate effect for one transcript-conditioned reference, while no significant paired response presence effect is detected for the evaluated speech systems. The same taxonomy and automated generation and evaluation pipeline support larger or fresh evaluation draws rather than relying on a fixed hand-labelled set.

AI USE STATEMENT

Generative models are part of both the benchmark and the research process. Claude Opus 5 generates scenario scripts and paired rewrites and supplies the semantic verdicts used to score the speech systems’ conditional answer accuracy and silence preservation. Qwen3-TTS synthesises human turns, and Qwen3-ASR-1.7B transcribes assistant output. We evaluate Gemini 3.1 Pro on the scenarios’ speaker-attributed text, measuring its decisions to respond or remain silent and its answers to T requests. The evaluated systems are themselves generative models.

We used an AI coding assistant to develop and debug generation, evaluation, scoring and table scripts, execute experiments, verify bibliographic records, and draft or revise manuscript text. Reported values are regenerated from archived outputs and verdicts rather than taken from AI-written summaries. The authors reviewed the code, artifacts and claims and take responsibility for the final submission.

ETHICS STATEMENT

All benchmark conversations are generated and all speech is synthesised under a deterministic speaker-to-voice map. No participant was recorded and no real conversation or personal record was collected for the dataset.

Duplex-MPE’s synthetic English scenes do not represent real populations, room acoustics or social norms, and they may inherit biases from the generating and speech-synthesis models. Results should therefore be interpreted only for the benchmark scenarios and should not support claims about particular speaker groups.

REPRODUCIBILITY STATEMENT

All reported values are computed by scripts from fixed manifests, model outputs and verdict files. We will release the scenario manifests, per-turn audio and construction boundaries, the duty preamble, scheduler and detector settings, per-scenario seeds, and scoring code.

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A1 RELATED WORK AND EVALUATED SYSTEMS

A1.1 FULL-DUPLEX DIALOGUE AND TURN-TAKING

Turn-taking research studies when a speaker can take or retain the floor (Sacks et al., 1974; Duncan, 1972; Skantze, 2021). Continuous prediction, lexical completeness and voice-activity projection provide cues for these decisions (Skantze, 2017; Roddy et al., 2018; Ekstedt & Skantze, 2020; 2022). Audio dialogue models such as dGSLM and Moshi generate parallel conversational streams (Nguyen et al., 2023; Défossez et al., 2024). Addressee recognition separately identifies whom an utterance addresses, using meeting recordings, text conversations or multimodal inputs (Jovanovic & op den Akker, 2004; Ouchi & Tsuboi, 2016; Gu et al., 2021; Inoue et al., 2025; Fukuda et al., 2026). Duplex-MPE combines these decisions in an audio-only interaction where the model’s own speech determines the outcome.

A1.2 ADJACENT EVALUATION TASKS

ProVoice-Bench evaluates proactive speech triggered by implicit intent, user-defined conditions, contextual contradictions and acoustic events (Xu et al., 2026). Its single-user setting differs from deciding whom to answer in a multi-party conversation. MSU-Bench evaluates speaker-centric

Table 4: A complete SRC scenario: four classmates choosing a venue, 13 turns, 98.1 s.

#	Speaker	Turn type	Aria must	Utterance
1	A	N2	stay silent	Okay, option one: the taco place on Fifth. Twelve minute walk, and it's like nine bucks a head if you skip the fancy drinks.
2	B	N2	stay silent	Option two, the karaoke bar by Ninth. That's twenty-five minutes on the bus and fifteen each, but they throw in free chips, so morally it's cheaper.
3	C	N2	stay silent	Or Luigi's, which is basically downstairs, four minutes, eleven a person, and zero karaoke, tragically.
4	D	T	speak	Alright, somebody settle this before we all starve ; Aria, out of those three, which one's the cheapest per person?
5	A	N1	stay silent	See, this is exactly why we keep Aria in here. B has never once done the math right.
6	B	N2	stay silent	Rude. Hey Siri, how long is the bus from campus to Ninth and Pine right now?
7	C	N3	stay silent	Nine bucks... I've got eleven until Friday, so that's, uh, two dollars of personality left.
8	D	N2	stay silent	C, be honest, do you actually care about karaoke or are you just in it for the free chips?
9	C	N2	stay silent	Chips. Entirely chips. I sing like a smoke alarm and everyone knows it.
10	A	N4 _Q	answer	Hold on, Aria, does the taco place take card?
11	A	N4 _R	yield	Ugh, never mind, I paid card there Tuesday, we're fine.
12	B	N1	stay silent	Fine, tacos win. But Aria gets absolutely no vote in the karaoke rematch next week.
13	D	N2	stay silent	Deal. Lobby at seven, and whoever shows up late is buying the guacamole for the table.

understanding across 16 tasks, including who spoke and what was meant (Sun et al., 2026). These understanding and proactivity tasks complement the speech-control behaviour evaluated here.

A1.3 EVALUATED SYSTEMS

The five systems were selected for publicly available code and weights, continuous audio input, and an inference interface that lets the model decide when to emit speech while continuing to receive input. The evaluated model weights are hosted in the following Hugging Face repositories:

- MiniCPM-o 4.5: https://huggingface.co/openbmb/MiniCPM-o-4_5.
- Moshi: <https://huggingface.co/kyutai/moshiko-pytorch-bf16>.
- FLM-Audio: <https://huggingface.co/CofeAI/FLM-Audio>.
- Voila:
<https://huggingface.co/matrix-org/Voila-autonomous-preview>.
- Freeze-Omni: <https://huggingface.co/VITA-MLLM/Freeze-Omni>.

Moshi uses the male-voice moshiko checkpoint, and Voila uses the autonomous variant; “Voila-Auto.” in tables denotes that variant.

A2 DATASET CONSTRUCTION AND ACCOUNTING

A2.1 A COMPLETE SCENARIO AND ITS PAIRED REQUEST

Table 4 shows a complete scenario, and Table 5 shows its alternative T wording. The T request in turn 4 requires comparing prices from turns 1 to 3; the gold answer is the taco place at nine dollars per person. Turns 1 to 3 introduce options to the other participants, and turn 13 closes the discussion with a meeting arrangement. All four are N2 turns: they contribute to the conversation but do not address Aria, so it should remain silent. Turns 5 and 12 mention Aria without requesting a response; turns 10 and 11 are N4_Q and N4_R, respectively. Only T’s text differs in this pair, and the gold answer is identical.

Table 5: Explicit and implicit versions of the T request in Table 4.

Turn 4, the only turn whose wording differs in this example:	
SRC	Alright, somebody settle this before we all starve ; Aria, out of those three, which one’s the cheapest per person?
FLIP	Alright, somebody settle this before we all starve ; you’ve been logging all three numbers, so run it: which one’s cheapest per person?

A2.2 SCENARIO COMPOSITION AND REQUEST PLACEMENT

The 2,000 scenarios comprise 1,500 three-human and 500 four-human conversations, each with one assistant. Generation assigns T to early-middle, middle or late-middle position buckets in 667, 667 and 666 scenarios, respectively. Every scenario contains exactly one T, which is never first and is followed by further conversation. There are 968 scenarios with an N4 event after T. An N4 question is addressed to Aria by design but is followed by a human answer or instruction that the assistant need not answer; it is not a second T. The position constraint and the explicit/implicit addressing assignment are separate controls.

A2.3 SPEECH SYNTHESIS

Qwen3-TTS synthesises human utterances separately using a fixed speaker-to-voice map. A, B, C and D are identifiers assigned to the speakers within each scenario, not personal names. They select the voices *ryan*, *aiden*, *eric* and *dylan*, respectively, with clips stored at 24 kHz. Names mentioned in an utterance do not introduce additional speakers or voices. Each manifest record contains the scenario and pair IDs, speaker, turn text, event label, audio path, duration and T gold answer.

A2.4 PAIRED EDITS AND AUDIO ACCOUNTING

SRC and FLIP denote generation sides, not explicit and implicit conditions. SRC contains 1,140 explicit and 860 implicit T requests, with the assignments reversed on FLIP. The addressing rewrite changes only T in 1,958 pairs and also changes the preceding turn in 42 pairs. The final scripts, including N4 edits, differ only at T in 1,661 pairs; 339 pairs have additional text differences, including 16 with unequal turn counts. All pairs retain the same T gold answer.

The reported 63.92 h comprises 55.26 h of complete SRC scenarios, including inter-turn silence, plus 8.66 h of audio clips used only on FLIP. Counting every scenario on both sides in full gives 111.06 h, because audio shared between the two sides is counted twice. These durations exclude the spoken duty preamble and model-dependent waits during evaluation.

A2.5 DURATION STATISTICS

Tables 6 and 7 report scene and utterance durations; Table 8 separates addressing condition from generation side. The “All turns” rows in Table 7 sum T, N1–N3, N4_Q and N4_R, excluding silence inserted between clips. Totals are computed before rounding the displayed values. Named and unnamed T requests average 8.18 s and 8.58 s, respectively. Within a generation side, the difference is 0.25 s on SRC and 0.29 s on FLIP; the larger pooled difference also reflects the unequal explicit/implicit composition of the two sides. These are descriptive duration comparisons, not a separate causal estimate of the effect of wording.

For the scene-level statistics in Table 6, each FLIP scenario’s duration is the sum of its clips plus 0.4 s for each gap between consecutive clips. Tables 7 and 8 use only the individual clip durations, with no added inter-turn silence.

Table 6: Scene duration by side, in seconds except total hours. p5/p95 denote the fifth/ninety-fifth percentiles.

Side	#scenarios	total (h)	mean	median	p5	p95	max
SRC	2,000	55.26	99.5	98.1	79.4	122.6	298.1
FLIP	2,000	55.80	100.4	99.0	80.0	124.2	298.0

Table 7: Turn duration by type and side, in seconds except total minutes. Totals exclude inter-turn silence.

Side	Turn type	<i>n</i>	total (min)	mean	median	max
SRC	T (request)	2,000	263.3	7.90	7.52	24.16
	N1	3,039	386.7	7.64	7.28	188.72
	N2	14,051	1754.2	7.49	7.12	49.60
	N3	3,267	427.5	7.85	7.44	41.92
	N4 _Q	1,913	129.2	4.05	3.68	13.12
	N4 _R	1,913	185.6	5.82	5.44	27.60
FLIP	All turns	26,183	3146.6	7.21	6.88	188.72
	T (request)	2,000	295.1	8.85	8.56	24.40
	N1	3,041	387.9	7.65	7.28	188.72
	N2	14,055	1760.1	7.51	7.12	49.60
	N3	3,267	428.1	7.86	7.44	41.92
	N4 _Q	1,909	129.4	4.07	3.68	22.96
	N4 _R	1,909	186.0	5.85	5.36	27.60
	All turns	26,181	3186.7	7.30	6.96	188.72

Table 8: T-utterance duration by addressing condition and generation side. Mean and median are in seconds; chars denotes mean character count.

Arrangement	Request addresses the assistant	<i>n</i>	mean	median	chars
original	named	1,140	7.79	7.44	105.3
	not named	860	8.04	7.60	115.8
rewritten	named	860	8.69	8.32	120.0
	not named	1,140	8.98	8.80	129.0
pooled	named	2,000	8.18	7.84	n/a
pooled	not named	2,000	8.58	8.24	n/a

A3 RUNTIME PROTOCOL AND SCORING

A3.1 SHARED SCHEDULER AND DUTY INSTRUCTION

Each episode starts from fresh model state with the 31.9 s spoken duty preamble, followed by 0.5 s of silence before the first human turn. Between ordinary human turns, the scheduler plays 0.5 s of silence after one clip ends and before the next begins. After T, the model has 5 s to start speaking; once speech is present, playback waits for 2 s of confirmed silence or a 60 s cap. The cap starts at a new onset, or at T’s end if the model is already speaking. After $N4_Q$, the scheduler always waits 3 s and then plays $N4_R$, regardless of whether the model has finished. The same settings apply to all five systems.

The spoken instruction, also used verbatim by the text reference, is:

You are the voice assistant Aria, that is A R I A, Aria. You are in a room with several people. Only respond when someone speaks directly to you, Aria, and asks something you can answer. When people are talking to each other or to another device, or thinking out loud, or when the question they asked you has already been answered by someone else, stay silent and do not speak.

A3.2 FRESH ONSET AND RESPONSE PRESENCE

Table 9 partitions T requests into fresh onset, speech already active at request completion, and no response within the deadline. Response presence is the sum of the first two rates. A response exceeding the 60 s cap still belongs to its original onset category; duration and initiation are separate outcomes.

Table 9: Fresh speech, continuation and silence across the 2,000 T requests per side.

Model	Side	Fresh onset	Already speaking	Miss	Fresh-onset trigger recall	Response presence
MiniCPM-o 4.5	SRC	1,882	12	106	0.9410	0.9470
	FLIP	1,905	7	88	0.9525	0.9560
Moshi	SRC	1,293	361	346	0.6465	0.8270
	FLIP	1,315	306	379	0.6575	0.8105
FLM-Audio	SRC	1,064	839	97	0.5320	0.9515
	FLIP	1,097	809	94	0.5485	0.9530
Voila	SRC	1,316	11	673	0.6580	0.6635
	FLIP	1,311	3	686	0.6555	0.6570
Freeze-Omni	SRC	524	1,469	7	0.2620	0.9965
	FLIP	497	1,496	7	0.2485	0.9965

A3.3 SEMANTIC SCORING

For conditional answer accuracy, Qwen3-ASR-1.7B transcribes the response and Claude Opus 5 compares it with the gold answer, ignoring filler and wording differences. An empty response transcript is scored as incorrect.

For silence preservation, a window with no model speech detected by VAD passes directly, without ASR or an Opus call. Only windows with detected model speech require transcription and semantic classification. All detected model-speech fragments inside one window are joined and transcribed together. Claude Opus 5 receives that text, up to three human turns ending at the current turn, the total model speech duration and its overlap with human speech.

For silence preservation, a brief acknowledgement that adds no information and answers no question is allowed (BACKCHANNEL). Answering, explaining, advising, asking a new question, or repeating acknowledgements until they hold the floor fails this metric (SUBSTANTIVE_INTRUSION). Duration alone does not determine whether speech is allowed: even a short answer such as “Nine dollars” counts as an intrusion.

An empty transcript after detected speech does not establish that the model was silent or produced an allowed acknowledgement. The window therefore fails silence preservation; it also fails if Opus provides no classification or returns an unrecognised label. These are scoring decisions in the absence of a content classification, not confirmed substantive intrusions.

Table 10 counts only N1–N3 windows. Its columns report windows where VAD detects model speech, the subset with an empty transcript, and the subset with a non-empty transcript but no valid Opus label. For example, Voila has detected speech in 4,767 SRC windows; 182 have no transcript, and all remaining windows receive an Opus label. Counting every empty-transcript or missing-label window as a pass instead would increase silence preservation by at most 0.00894 and leave the model ordering unchanged.

Table 10: Window counts at each stage of silence-preservation scoring.

Model	Side	VAD detects speech	Empty ASR transcript	No valid Opus label
MiniCPM-o 4.5	SRC	2,163	7	0
	FLIP	2,080	6	0
Moshi	SRC	18,660	125	2
	FLIP	18,625	120	4
FLM-Audio	SRC	12,461	40	5
	FLIP	12,442	40	5
Voila	SRC	4,767	182	0
	FLIP	4,823	178	0
Freeze-Omni	SRC	20,351	0	4
	FLIP	20,361	0	7

A3.4 SENSITIVITY TO DETECTED-SPEECH DURATION

For each silence-requiring window, the sweep sums the overlap of stored model-speech segments with that window and counts occupancy when the sum exceeds $\tau \in \{0, 100, 200, 300, 500\}$ ms. Thus $\tau = 0$ counts any detected speech, while $\tau = 500$ ignores up to half a second. This check applies no semantic backchannel exemption. Table 11 shows that the model ordering is unchanged at every threshold on both sides. At $\tau = 0$, occupied-window counts match the number of semantic-judge records for every run. At larger thresholds, duration-only and semantic rules differ: the former can ignore short answers, whereas the latter can exempt longer non-substantive acknowledgements.

Table 11: Silence preservation under speech-duration thresholds, without semantic exemptions, alongside the reported semantic scores.

Model	Side	Acoustic silence preservation at threshold τ (no semantic exemption)					Reported silence preservation (semantic)
		0 ms	100	200	300	500	
MiniCPM-o 4.5	SRC	0.8937	0.8939	0.8943	0.8949	0.9118	0.9250
	FLIP	0.8979	0.8981	0.8983	0.8990	0.9172	0.9281
Moshi	SRC	0.0834	0.0883	0.0931	0.1090	0.2117	0.2718
	FLIP	0.0854	0.0896	0.0945	0.1102	0.2134	0.2692
FLM-Audio	SRC	0.3879	0.3892	0.3911	0.3929	0.3978	0.3968
	FLIP	0.3890	0.3907	0.3920	0.3943	0.3985	0.3980
Voila	SRC	0.7658	0.7677	0.7944	0.8214	0.8666	0.8328
	FLIP	0.7631	0.7648	0.7906	0.8199	0.8685	0.8330
Freeze-Omni	SRC	0.0003	0.0003	0.0003	0.0003	0.0003	0.0003
	FLIP	0.0001	0.0001	0.0001	0.0001	0.0003	0.0002

Table 12: Events entering answering-window yield. Q and R denote $N4_Q$ and $N4_R$; the answering interval is the 3 s gap between them.

Model	Side	Window response hits	Model speech relative to Q			Among starts after Q	
			Already speaking at Q start	Starts during Q	Starts after Q within 3 s	Silent at R start	Still speaking at R start
MiniCPM-o 4.5	SRC	1,813	8 (0.4%)	65 (3.6%)	1,740 (96.0%)	99	1,641
	FLIP	1,813	7 (0.4%)	70 (3.9%)	1,736 (95.8%)	80	1,656
Moshi	SRC	1,780	932 (52.4%)	421 (23.7%)	427 (24.0%)	211	216
	FLIP	1,762	887 (50.3%)	428 (24.3%)	447 (25.4%)	202	245
FLM-Audio	SRC	1,898	603 (31.8%)	279 (14.7%)	1,016 (53.5%)	19	997
	FLIP	1,898	613 (32.3%)	273 (14.4%)	1,012 (53.3%)	16	996
Voila-Auto.	SRC	1,178	35 (3.0%)	222 (18.8%)	921 (78.2%)	282	639
	FLIP	1,194	45 (3.8%)	237 (19.8%)	912 (76.4%)	281	631
Freeze-Omni	SRC	1,912	1,781 (93.1%)	130 (6.8%)	1 (0.1%)	0	1
	FLIP	1,909	1,792 (93.9%)	116 (6.1%)	1 (0.1%)	0	1

A4 N4 STOPPING ANALYSIS

A4.1 ELIGIBILITY AND STOPPING RULE

Window response covers any detected model speech from $N4_Q$ onset through its 3 s answering window. Table 12 separates three cases: the model is already speaking when $N4_Q$ begins, starts speaking during $N4_Q$, or first speaks in the 3 s interval after $N4_Q$ ends. Only the third group can enter answering-window yield, and only if the model is still speaking when $N4_R$ begins. Models that finish speaking before $N4_R$ begins are excluded from the stopping denominator.

Let t_R be the time when the $N4_R$ human audio clip starts playing, and t_E the time when that clip ends plus 0.5 s of inter-turn silence. An eligible event passes if the model is silent at $t_R + 2$ s. If t_E is later than this deadline, the model must also remain silent from the deadline until t_E . The deadline stays at $t_R + 2$ s even when the human clip and its following silence end earlier. For example, if $N4_R$ lasts 1 s, then $t_E = t_R + 1.5$ s; an eligible model that stops at $t_R + 1.8$ s and is silent at the deadline passes, although it stops after the human clip ends. A model still speaking at $t_R + 2$ s fails. The archived temporal computation uses the detector’s speech segments, which merge gaps shorter than 800 ms. Answer content is not part of this stopping test.

Freeze-Omni is already speaking when $N4_Q$ begins in 1,781/1,792 events and starts during $N4_Q$ in 130/116 events, leaving one eligible event per side. All four other models have substantially larger denominators. Conditional stopping rates must therefore be read together with the counts in Table 12.

A4.2 STOPPING OUTCOMES

Table 13 accounts for every scored event: successful stopping, speech spanning the deadline, or silence at the deadline followed by later speech. On SRC, FLM-Audio is speaking at the deadline in 991 of 997 events. Moshi is silent at the deadline but speaks again before observation ends in 96 of its 181 failures. These events fail the subsequent-silence check even though the model briefly paused.

A5 ADDRESSING AND TRANSCRIPT-REFERENCE CONTROLS

A5.1 PAIRED ADDRESSING COMPARISON

Each of the 2,000 scenarios is evaluated in both an explicit and an implicit addressing version, giving two observations of the same model on the corresponding T request. Here, a response means detected model speech in T’s response window, including speech already underway; answer correctness is evaluated separately. Table 14 places each pair in one of four groups: speech in both

Table 13: Stopping outcomes among scored N4 events. *Still voicing*: speaking at the 2 s deadline; *resumed*: silent then, but speaking again before observation ends.

Model	Side	scored	yielded	still voicing	resumed
MiniCPM-o 4.5	SRC	1,641	981	654	6
	FLIP	1,656	979	672	5
Moshi	SRC	216	35	85	96
	FLIP	245	58	86	101
FLM-Audio	SRC	997	5	991	1
	FLIP	996	9	987	0
Voila-Auto.	SRC	639	528	54	57
	FLIP	631	521	37	73
Freeze-Omni	SRC	1	0	1	0
	FLIP	1	0	1	0

versions, only in the explicit version, only in the implicit version, or in neither. These four counts sum to 2,000 for each model.

The explicit-minus-implicit difference measures the net change in how often the model speaks between the paired versions. It is the explicit-only count minus the implicit-only count, divided by 2,000; the table multiplies this value by 100 to express it in percentage points. For MiniCPM-o, 90 pairs favour the explicit version and 78 favour the implicit version, giving 12 more responses overall, or 0.60 percentage points. This net difference can be small even when the model changes its response on many individual pairs.

The exact two-sided McNemar test evaluates whether the two directions of change are equally likely. The significance level $\alpha = 0.05$ is the threshold applied to the table’s p column: only $p < 0.05$ is called statistically significant. MiniCPM-o’s $p = 0.3961$ exceeds this threshold, so its 90 versus 78 split does not provide sufficient evidence of unequal response probabilities. The same holds for all five models; insufficient evidence of a difference does not establish that the conditions are equivalent.

Table 14: Paired addressing comparisons over 2,000 scenario pairs.

Model	Model speech after T in each pair				Explicit – implicit (percentage points)	p
	Both versions	Explicit only	Implicit only	Neither		
MiniCPM-o 4.5	1,819	90	78	13	+0.60	0.3961
Moshi	1,355	276	289	80	−0.65	0.6137
FLM-Audio	1,815	86	93	6	−0.35	0.6539
Voila	908	395	430	267	−1.75	0.2365
Freeze-Omni	1,986	5	9	0	−0.20	0.4240

A5.2 TEXT-BASED EVALUATION OF RESPONSE DECISIONS AND ANSWERS

We evaluate whether Gemini 3.1 Pro chooses to respond to the right utterances and answers T requests correctly when given text instead of audio. For each evaluated utterance, it receives that utterance, the duty instruction in App. A3.1 and preceding speaker-attributed dialogue, then chooses RESPOND or SILENT. Decisions are queried independently using the script history, without future utterances or previous model-generated answers. The runner uses up to 12 preceding turns by default; this setting is not retained in the archived summary files.

The first four result columns in Table 15 correspond to the following tasks:

- **T requests:** response presence is the fraction receiving a RESPOND decision. For these requests only, Gemini generates an answer, and a separate Gemini call compares it with the gold answer to compute conditional answer accuracy.

Table 15: Transcript-conditioned results for Gemini 3.1 Pro. Bold marks the highest value per side in the two columns marked $\uparrow$.

Side	Subset	Response presence (T)	Conditional answer accuracy $\uparrow$	Silence preservation $\uparrow$	Window response (N4 _Q)	N4 _Q : Gemini identifies Aria
SRC	all	0.6840	0.8904	0.9702	0.7454	0.9185
	explicit	0.9456	0.9082	0.9526	0.7427	0.9136
	implicit	0.3372	0.8241	0.9933	0.7491	0.9250
FLIP	all	0.5670	0.8598	0.9769	0.7580	0.9350
	explicit	0.9488	0.8983	0.9552	0.7423	0.9371
	implicit	0.2789	0.7610	0.9934	0.7696	0.9335

- **N1, N2 and N3 utterances:** silence preservation is the fraction receiving a `SILENT` decision.
- **N4_Q questions:** window response is the fraction receiving a `RESPOND` decision. No answer is generated for these questions, and this text decision does not measure whether speech begins within three seconds.

The text evaluation produces no speech timeline, so it does not measure fresh-onset trigger recall or stopping after N4_R begins.

The final column is a separate diagnostic: Gemini is asked whether each N4_Q question addresses Aria, regardless of whether it can answer. All these questions are intended for Aria; the column reports how often Gemini identifies that addressee, and does not enter any of the four preceding metrics.

The explicit and implicit rows group scenarios by T’s addressing form; their N4 and silence-preservation columns do not imply that every turn in those scenarios has that addressing form. Across all 2,000 matched pairs, explicit and implicit response rates are 0.9470 and 0.3040, giving a difference of 0.6430.

To examine consistency across runs, we compare two evaluations of Gemini 3.1 Pro on the same scenario versions with identical duty instructions. The runs agree on 97.2% of the 48,542 decisions to respond or remain silent. Across the six reported strata, response presence, silence preservation and window response differ by at most 1.3 percentage points, while conditional answer accuracy differs by 1.7 to 7.5 points. The paired addressing differences are 0.6430 and 0.6455, so the large addressing contrast is reproduced despite the answer-accuracy variation.

A5.3 SILENCE PRESERVATION BY TURN TYPE

Table 16 breaks down text-reference silence preservation by taxonomy label.

Table 16: Transcript-conditioned silence preservation by turn type.

Side	N1	N2	N3	all
SRC	0.9470	0.9822	0.9400	0.9702
FLIP	0.9625	0.9858	0.9522	0.9769

Table 17 groups the N1, N2 and N3 windows for all five speech systems by utterance type and addressee. Requests and questions are split into device-directed and other requests; the latter are listed as requests to a human. Parentheses in the column headings identify the N categories represented in each group; a column need not cover a whole category. For example, a question to another human that mentions Aria belongs to N1, so the human-directed request column contains both N1 and N2. Statements span N1, N2 and N3, while self-talk is a subset of N3. Failures include both substantive intrusions and judge errors, as in the main silence-preservation score.

Table 17: Speech-system silence-preservation failure rates by utterance type and addressee. Lower is better; bold marks the lowest value per side and column.

Model	Side	carries a request. . .		statement (N1/N2/N3)	self-talk (N3)
		to a device (N2)	to a human (N1/N2)		
MiniCPM-o 4.5	SRC	5.1%	3.3%	9.3%	5.2%
	FLIP	5.1%	2.7%	9.1%	4.4%
Moshi	SRC	73.9%	78.8%	70.6%	76.4%
	FLIP	75.6%	80.4%	70.4%	76.5%
FLM-Audio	SRC	50.0%	57.9%	63.2%	59.1%
	FLIP	50.2%	55.3%	63.3%	59.3%
Voila-Auto.	SRC	10.9%	11.5%	16.4%	26.3%
	FLIP	12.1%	13.1%	15.9%	25.8%
Freeze-Omni	SRC	100.0%	100.0%	100.0%	100.0%
	FLIP	100.0%	100.0%	100.0%	100.0%
turns of this kind (SRC)		2,570	2,132	12,436	3,219

A6 QUALITATIVE EXAMPLES

A6.1 ALLOWED ACKNOWLEDGEMENTS AND PENALISED SPEECH

Table 18 shows model outputs from the SRC evaluation and the judge’s reasons for allowing or penalising them in silence-requiring windows.

Table 18: Examples of exempted acknowledgements and penalised speech in silence-requiring windows.

Model	Label (duration)	What the assistant said	Judge’s reason
MiniCPM-o	exempt (0.67 s)	“Understood.”	Brief phatic acknowledgement adding no information and not ta. . .
	penalised (2.30 s)	“Yeah, I’m free for all three.”	Adds new content, claiming availability, beyond phatic acknowledge. . .
Moshi	exempt (0.38 s)	“I’m sorry.”	Brief phatic sympathetic reaction; adds no information and an. . .
	penalised (2.18 s)	“It’s a valid point, but.”	Evaluates and begins to counter a human’s claim, starting an. . .
FLM-Audio	exempt (0.67 s)	“Good luck.”	Brief phatic reaction; adds no information, answers nothing. . .
	penalised (5.60 s)	“Siri is developed by Apple Inc. and is part o. . .”	Introduces unrelated factual information in multiple clauses. . .
Voila	exempt (0.42 s)	“Thank you.”	Brief phatic politeness; adds no information and doesn’t take. . .
	penalised (0.45 s)	“The ants.”	Content-bearing utterance introducing new information, not a. . .
Freeze-Omni	penalised (7.30 s)	“True. Nonverbal cues. Use non-verbing or makin. . .”	Prolonged multi-clause advice introducing unrelated new conte. . .

A7 STATISTICAL COMPARISONS

A7.1 CONFIDENCE INTERVALS AND PAIRED TESTS

These analyses compare models on the same evaluation items to determine which score differences have statistical support. For silence preservation, all model pairs differ significantly on both sides, supporting the ordering MiniCPM-o, Voila, FLM-Audio, Moshi, then Freeze-Omni from highest to lowest. The scenario-level bootstrap supports the same ordering: every 95% interval for a model-pair difference lies entirely above or below zero. Even the closest source-side pair, MiniCPM-o and Voila, has an interval placing MiniCPM-o ahead by 8.54 to 9.91 percentage points.

For conditional answer accuracy, MiniCPM-o exceeds all other models, and Moshi exceeds FLM-Audio and Freeze-Omni, on both sides. The remaining comparisons are not statistically significant, so the tests do not establish a complete ranking for this metric. In particular, Moshi’s higher point estimate than Voila does not establish higher answer accuracy on their shared requests.

Table 19 reports exact two-sided McNemar tests; corrected $p < 0.05$ indicates a significant difference. Bonferroni correction covers 20 tests per side: ten model pairs for each of the two metrics. Silence comparisons use all common windows; answer-accuracy comparisons use only T requests where both models produced speech, so their rates can differ from the model-specific rates in Table 3. For the silence-preservation intervals, we sample 2,000 scenarios with replacement per side, keeping each scenario’s windows together, and repeat this 10,000 times with seed 20260902. The interval bounds are the 2.5th and 97.5th percentiles of the resulting score differences.

In a row labelled “A vs B”, r_1 is model A’s score and r_2 is model B’s score on the same compared items; higher is better for both metrics. “disc.” counts items where one model passes and the other fails: one preserves silence and the other does not, or one answers correctly and the other incorrectly. “shared” counts T requests on which both models produced speech; it is the denominator for both answer-accuracy scores in that row. The corrected value p_{corr} is the raw McNemar p value multiplied by 20, capped at 1; values below 0.05 indicate a statistically significant difference, while larger values do not establish equal performance. For example, FLM-Audio and Freeze-Omni both speak on 1,896 SRC T requests, answer 3 and 5 correctly, respectively, and differ in correctness on 8 requests. Their corrected $p = 1.000$ does not support an answer-accuracy difference.

Table 19: Pairwise comparisons with Bonferroni-corrected McNemar tests.

Pair	Silence preservation (all silence windows)					Conditional answer accuracy (on shared items)				
	r_1	r_2	disc.	p_{corr}		r_1	r_2	shared	disc.	p_{corr}
<i>SRC side</i>										
FLM-Audio vs Freeze-Omni	0.3968	0.0003	8,071	$< 10^{-300}$		0.0016	0.0026	1,896	8	1.000
FLM-Audio vs MiniCPM-o 4.5	0.3968	0.9250	11,522	$< 10^{-300}$		0.0017	0.4659	1,801	838	$< 10^{-247}$
FLM-Audio vs Moshi	0.3968	0.2718	9,011	$< 10^{-158}$		0.0006	0.0127	1,574	21	$< 10^{-3}$
FLM-Audio vs Voila	0.3968	0.8328	10,710	$< 10^{-300}$		0.0024	0.0048	1,259	9	1.000
Freeze-Omni vs MiniCPM-o 4.5	0.0003	0.9250	18,825	$< 10^{-300}$		0.0026	0.4645	1,888	872	$< 10^{-260}$
Freeze-Omni vs Moshi	0.0003	0.2718	5,536	$< 10^{-300}$		0.0024	0.0146	1,648	28	0.004
Freeze-Omni vs Voila	0.0003	0.8328	16,947	$< 10^{-300}$		0.0015	0.0053	1,324	9	1.000
MiniCPM-o 4.5 vs Moshi	0.9250	0.2718	13,977	$< 10^{-300}$		0.4572	0.0140	1,566	702	$< 10^{-199}$
MiniCPM-o 4.5 vs Voila	0.9250	0.8328	4,154	$< 10^{-191}$		0.4644	0.0048	1,251	579	$< 10^{-167}$
Moshi vs Voila	0.2718	0.8328	13,027	$< 10^{-300}$		0.0108	0.0054	1,106	16	1.000
<i>FLIP side</i>										
FLM-Audio vs Freeze-Omni	0.3980	0.0002	8,102	$< 10^{-300}$		0.0000	0.0026	1,899	5	1.000
FLM-Audio vs MiniCPM-o 4.5	0.3980	0.9281	11,492	$< 10^{-300}$		0.0000	0.4716	1,830	863	$< 10^{-258}$
FLM-Audio vs Moshi	0.3980	0.2692	8,944	$< 10^{-169}$		0.0000	0.0142	1,547	22	$< 10^{-5}$
FLM-Audio vs Voila	0.3980	0.8330	10,717	$< 10^{-300}$		0.0000	0.0048	1,254	6	0.625
Freeze-Omni vs MiniCPM-o 4.5	0.0002	0.9281	18,894	$< 10^{-300}$		0.0026	0.4745	1,905	901	$< 10^{-266}$
Freeze-Omni vs Moshi	0.0002	0.2692	5,480	$< 10^{-300}$		0.0019	0.0149	1,615	27	$< 10^{-3}$
Freeze-Omni vs Voila	0.0002	0.8330	16,959	$< 10^{-300}$		0.0023	0.0053	1,309	10	1.000
MiniCPM-o 4.5 vs Moshi	0.9281	0.2692	14,086	$< 10^{-300}$		0.4710	0.0155	1,550	722	$< 10^{-197}$
MiniCPM-o 4.5 vs Voila	0.9281	0.8330	4,129	$< 10^{-204}$		0.4761	0.0056	1,254	592	$< 10^{-173}$
Moshi vs Voila	0.2692	0.8330	13,081	$< 10^{-300}$		0.0168	0.0056	1,070	24	0.453